

AdaBoost

M514: Econometrics II

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Outline

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References

- ▶ Trevor Hastie, Robert Tibshirani, Jerome Friedman (2009), "The Elements of Statistical Learning (ESL)", Chapter 5.1 ~ 5.3, 10.1 ~ 10.6

AdaBoost

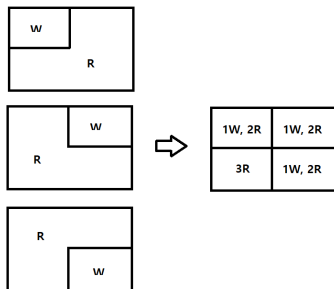
Given,

$$\{(x_i, y_i)\}_{i=1}^n, \quad x_i \in \mathbb{R}^p, \quad y_i \in \{-1, 1\}$$

we want to estimate a classifier $G : \mathbb{R}^p \rightarrow \{-1, 1\}$ such that $G(x_{n+1})$ is a good predictor for y_{n+1} .

- ▶ A **weak classifier** is one whose error rate is only slightly better than random guessing.
- ▶ The purpose of boosting is to sequentially apply the weak classification algorithm to repeatedly modified versions of the data, thereby producing a sequence of weak classifiers $G_m(x), m = 1, 2, \dots, M$.

AdaBoost



Suppose we have three weak classifiers trained on the same sample: G_1, G_2, G_3 each with 75% accuracy. Under ideal circumstances, we can combine these weak classifiers to create a "strong" classifier with 100% accuracy with simple majority vote rule. Boosting is a simple extension in which we use different weights to each samples and to each weak classifiers.

AdaBoost

AdaBoost Algorithm

1. Initialize weights given to each samples: $w_i = 1/n$
2. For $m = 1$ to M ,
 - (a) Fit a classifier $G_m(x)$ to the training data using weights w_i .
 - (b) Compute the error rate for the model m

$$\text{err}_m = \frac{\sum_{i=1}^n w_i 1_{\{y_i \neq G_m(x_i)\}}}{\sum_{i=1}^n w_i}$$

- (c) Compute $\alpha_m = \log((1 - \text{err}_m)/\text{err}_m)$
 - (d) Set $w_i \leftarrow w_i \cdot \exp(\alpha_m \cdot 1_{\{y_i \neq G_m(x_i)\}})$, $i = 1, \dots, n$
 3. Output $G(x) = \text{sign} \left(\sum_{m=1}^M \alpha_m G_m(x) \right)$.
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Features of AdaBoost algorithm

- ▶ At step m , observations that were misclassified by the classifier $G_{m-1}(x)$ have their weights increased. (Line 2d)
- ▶ Classifiers with smaller weighted error rates are given more weights in producing the final classifier $G(x)$. (Line 2c, Line 3).

AdaBoost

- ▶ Note that in the algorithm above, we have not specified how the classifier G_m is estimated (Line 2a). AdaBoost can be applied to any classifiers. However, in most applications, AdaBoost is applied to decision trees. It can be argued that decision trees are ideal base learner for boosting (Hastie et al. 2009, Chapter 10.7 and 10.9).
- ▶ In the sections below, we will show that AdaBoost can be seen as fitting a basis expansion of x using forward stagewise modeling.

Basis expansion

In order to incorporate nonlinearity, it is common to augment or replace vector of inputs $x \in \mathbb{R}^p$ with nonlinear transformations of x .

Let $b_m(x) : \mathbb{R}^p \rightarrow \mathbb{R}, m = 1, \dots, M$ be transformations of x . We model the conditional expectation as

$$f(x) = \sum_{m=1}^M \beta_m b_m(x)$$

Note that $f(x)$ is linear in β_m , but no longer linear in x .

Basis expansion

If we know which basis functions to use, we may estimate the model by minimizing a loss function.

$$\min_{\{\beta_m\}_{m=1}^M} \sum_{i=1}^n L \left(y_i, \sum_{m=1}^M \beta_m b_m(x_i) \right)$$

In case of squared error loss,

$$\min_{\{\beta_m\}_{m=1}^M} \sum_{i=1}^n \left(y_i - \sum_{m=1}^M \beta_m b_m(x_i) \right)^2$$

which is just ordinary least squares of y_i on $(b_1(x_i), \dots, b_M(x_i))'$.

Basis expansion

Suppose the basis functions $b_m(x)$, $m = 1, \dots, M$ are characterized by parameter γ_m :

$$b_m(x) = b(x; \gamma_m)$$

If the basis functions are not predetermined and need to be estimated, we may estimate $\{\beta_m, \gamma_m\}_{m=1}^M$ by jointly minimizing a loss function.

$$\min_{\{\beta_m, \gamma_m\}_{m=1}^M} \sum_{i=1}^n L \left(y_i, \sum_{m=1}^M \beta_m b(x_i, \gamma_m) \right) \quad (1)$$

Solving the joint minimization problem is computationally intensive. Alternatively, we can find an approximation to the solution of (1) by **forward stagewise modeling**.

Forward stagewise modeling

Forward Stagewise Modeling

1. Initialize $f_0(x) = 0$.
2. For $m = 1$ to M
 - (a) Compute

$$(\beta_m, \gamma_m) = \arg \min_{\beta, \gamma} \sum_{i=1}^n L(y_i, f_{m-1}(x_i) + \beta b(x_i; \gamma))$$

- (b) Set $f_m(x) = f_{m-1}(x) + \beta_m b(x; \gamma_m)$
-

Forward stagewise modeling approximates the solution to (1) by sequentially adding new basis functions to the expansion without adjusting the parameters and coefficients of those that have already been added.

Forward stagewise modeling

In case of squared error loss, we seek to minimize

$$\begin{aligned}L(y_i, f_{m-1} + \beta b(x; \gamma)) &= (y_i - f_{m-1} - \beta b(x_i; \gamma))^2 \\ &= (r_{im} - \beta b(x_i; \gamma))^2\end{aligned}$$

at each iteration, where $r_{im} = y_i - f_{m-1}(x_i)$ is simply the residual of the current model. Thus, for squared error loss, the term $\beta_m b(x; \gamma_m)$ that best fits the current residuals is added to the expansion at each step.

Forward stagewise modeling: AdaBoost

Recall that

$$G(x) = \text{sign} \left(\sum_{m=1}^M \alpha_m G_m(x) \right)$$

It can be shown that implementing AdaBoost can be seen as fitting a basis expansion on x , where the basis functions are the individual classifiers $G_m(x) \in \{-1, 1\}$. Specifically, fitting equation (1) using $G_m(x)$ as basis function with exponential loss and forward stagewise modeling is same as AdaBoost.

$$\min_{\{\beta_m, G_m\}_{m=1}^M} \sum_{i=1}^n L \left(y_i, \sum_{m=1}^M \beta_m G_m(x_i) \right)$$

Where exponential loss function is:

$$L(y, G(x)) = \exp(-yG(x))$$

Forward stagewise modeling: AdaBoost

With $G_m : \mathbb{R}^p \rightarrow \{-1, 1\}$ as a basis function, forward stagewise modeling seeks to solve

$$\begin{aligned}(\beta_m, G_m) &= \arg \min_{\beta, G} \sum_{i=1}^n L(-y_i, f_{m-1}(x_i) + \beta G(x_i)) \\ &= \arg \min_{\beta, G} \sum_{i=1}^n \exp(-y_i(f_{m-1}(x_i) + \beta G(x_i)))\end{aligned}$$

at each iteration (Line 2a, Forward Stagewise Modeling). Where the second equality comes from exponential loss.

Forward stagewise modeling: AdaBoost

Also,

$$\begin{aligned}(\beta_m, G_m) &= \arg \min_{\beta, G} \sum_{i=1}^n \exp(-y_i(f_{m-1}(x_i) + \beta G(x_i))) \\&= \arg \min_{\beta, G} \sum_{i=1}^n e^{-y_i f_{m-1}(x_i)} \cdot \exp(-y_i \beta G(x_i)) \\&= \arg \min_{\beta, G} \sum_{i=1}^n w_i^{(m)} \cdot \exp(-y_i \beta G(x_i))\end{aligned}\tag{2}$$

where $w_i^{(m)} = e^{-y_i f_{m-1}(x_i)}$. Note that since $w_i^{(m)}$ does not depend on β or $G(x)$, it can be regarded as weights applied to each sample at m -th iteration.

Forward stagewise modeling: AdaBoost

We can first solve for $G(x)$ in (2) by noting that for any value of β , solution to for $G(x)$ is

$$G_m(x) = \arg \min_G \sum_{i=1}^n w_i^{(m)} 1_{\{y_i \neq G(x_i)\}} \quad (3)$$

which is a classifier that minimizes weighted error rate. To see this, note that the criterion in (2) can be written as

$$\begin{aligned} & \sum_{i=1}^n w_i^{(m)} \cdot \exp(-y_i \beta G(x_i)) \\ &= \sum_{i=1}^n w_i^{(m)} \cdot \left(e^{-\beta} 1_{\{y_i = G(x_i)\}} + e^{\beta} 1_{\{y_i \neq G(x_i)\}} \right) \\ &= (e^{\beta} - e^{-\beta}) \sum_{i=1}^n w_i^{(m)} 1_{\{y_i \neq G(x_i)\}} + e^{-\beta} \sum_{i=1}^n w_i^{(m)} \end{aligned} \quad (4)$$

Forward stagewise modeling: AdaBoost

For any $G(x)$ in (4), we can easily solve for β . The FOC is given by

$$\begin{aligned}(e^\beta + e^{-\beta}) \sum_{i=1}^n w_i^{(m)} 1_{\{y_i \neq G(x_i)\}} - e^{-\beta} \sum_{i=1}^n w_i^{(m)} &= 0 \\ \implies (e^{2\beta} + 1) \sum_{i=1}^n w_i^{(m)} 1_{\{y_i \neq G(x_i)\}} &= \sum_{i=1}^n w_i^{(m)} \\ \implies e^{2\beta} &= \frac{\sum_{i=1}^n w_i^{(m)} - \sum_{i=1}^n w_i^{(m)} 1_{\{y_i \neq G(x_i)\}}}{\sum_{i=1}^n w_i^{(m)} 1_{\{y_i \neq G(x_i)\}}}\end{aligned}$$

Thus, the solution is given by

$$\beta_m = \frac{1}{2} \log \frac{1 - \text{err}_m}{\text{err}_m}$$

where

$$\text{err}_m = \frac{\sum_{i=1}^n w_i^{(m)} 1_{\{y_i \neq G_m(x_i)\}}}{\sum_{i=1}^n w_i^{(m)}}$$

Forward stagewise modeling: AdaBoost

Comparing the solution

$$\beta_m = \frac{1}{2} \log \frac{1 - \text{err}_m}{\text{err}_m}, \quad \text{err}_m = \frac{\sum_{i=1}^n w_i^{(m)} 1_{\{y_i \neq G_m(x_i)\}}}{\sum_{i=1}^n w_i^{(m)}}$$

we see that α_m in the (Line 2c, AdaBoost Algorithm) is same as $2\beta_m$. Multiplying by 2 to all the basis function does not have any affect on the final classifier.

Forward stagewise modeling: AdaBoost

After computing β_m and $G_m(x)$, we update $f_m(x)$ (Line 2b, Forward Stagewise Modeling), (Line 3, AdaBoost Algorithm):

$$f_m = f_{m-1}(x) + \beta_m G_m(x)$$

which causes the weights for the next iteration to be:

$$\begin{aligned} w_i^{(m+1)} &= w_i^{(m)} \cdot e^{-\beta_m y_i G_m(x_i)} \\ &= w_i^{(m)} \cdot e^{\alpha_m 1_{\{y_i \neq G_m(x_i)\}}} \cdot e^{-\beta_m} \end{aligned} \tag{5}$$

where $-y_i G_m(x_i) = 21_{\{y_i \neq G_m(x_i)\}} - 1$ and $\alpha_m = 2\beta_m$. Since the factor $e^{-\beta_m}$ in (5) multiplies all weights by the same value, it has no effect. Thus, (5) is same as (Line 2d, AdaBoost Algorithm).

Forward stagewise modeling: AdaBoost

Finally, one can view (Line 2a, AdaBoost Algorithm) as approximately solving (3). Thus, we conclude that AdaBoost can be seen as fitting a model with G_m as basis function using forward stagewise modeling to minimize exponential loss function. This approach to understanding boosting methods as a way of minimizing a loss function is useful in that it can lead to deeper understanding of other boosting methods (Gradient Boosting, XGBoost).

Loss function

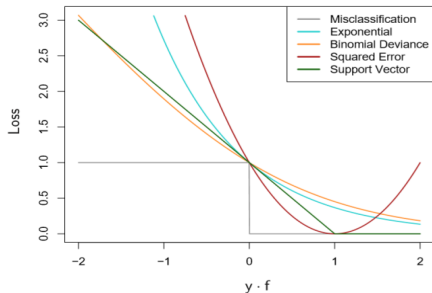


FIGURE 10.4. Loss functions for two-class classification. The response is $y = \pm 1$; the prediction is f , with class prediction $\text{sign}(f)$. The losses are misclassification: $I(\text{sign}(f) \neq y)$; exponential: $\exp(-yf)$; binomial deviance: $\log(1 + \exp(-2yf))$; squared error: $(y - f)^2$; and support vector: $(1 - yf)_+$ (see Section 12.3). Each function has been scaled so that it passes through the point $(0, 1)$.

The exponential loss function looks similar to Binomial Deviance loss function. However, it is less robust to outliers.